

BLG 317E - DATABASE SYSTEMS

FALL 2024 (CRN: 13591)

PROJECT PROPOSAL

Note that your selected dataset should be extensive enough to allow at least 5 main tables, and each table must have at least 3 non-key columns. Each team member must be responsible for at least one main table. Do your best in identifying table requirements. You will be asked to change your selected dataset if it does not seem to meet the minimum requirements. Please see the project evaluation and project self-assessment forms before finalizing your dataset to get an idea about the project evaluation criteria and process.

Project Group Name	Medico
Dataset title	Hospital Management Dataset
Dataset URL	https://www.kaggle.com/datasets/rajkumarpadmanabhan/ca-hospital-dataset-q1-2025
Data description	1) patients.csv — Patient demographics & enrollment - MAIN PK: patient_id FK: insurance_type → insurers.code non-key columns : first_name, last_name, dob, age, gender, ethnicity, insurance_type, marital_status, address, city, state, zip, phone, email, registration_date Represents the FHIR Patient concept; demographics are stable reference data joined by patient_id. 2) providers.csv — Clinicians & identifiers - MAIN PK: provider_id FK: head_id → specialty_heads.head_id Non-key: name, department, specialty, npi, inhouse, location, years_experience, contact_info, email npi is the U.S. National Provider Identifier (10 digits), the standard identifier in HIPAA transactions. 3) encounters.csv — Visits/episodes of care - MAIN PK: encounter_id FK: patient_id → patients.patient_id

- provider_id → **providers.provider_id**

Non-key: visit_date, visit_type, department, reason_for_visit, diagnosis_code, admission_type, discharge_date, length_of_stay, status, readmitted_flag

A FHIR Encounter: the actual interaction between a patient and provider(s). Use patient_id/provider_id to link.

4) diagnoses.csv — Diagnostic coding per encounter - MAIN

PK: diagnosis_id

FK: encounter_id → **encounters.encounter_id**

Non-key: diagnosis_code, diagnosis_description, primary_flag, chronic_flag

Diagnosis_code is intended for ICD-10/ICD-10-CM coding of conditions; primary_flag denotes the principal diagnosis.

5) procedures.csv — Procedures/services rendered - MAIN

PK: procedure_id

FK:

- encounter_id → **encounters.encounter_id**
- provider_id → **providers.provider_id**

Non-key: procedure_code, procedure_description, procedure_date, procedure_cost

Procedure_code is typically CPT (or HCPCS) for billing/reporting of services.

6) lab_tests.csv — Laboratory tests & results - MAIN

PK: lab_id

FK: encounter_id → **encounters.encounter_id**

Non-key: test_name, test_code, specimen_type, test_result, units, normal_range, test_date, status

test_code is designed for LOINC, the standard coding for lab orders/results; each record is an individual lab observation.

7) medications.csv — Medications prescribed/administered - MAIN

PK: medication_id

FK:

- encounter_id → **encounters.encounter_id**

- prescriber_id → **providers.provider_id**

Non-key: drug_name, dosage, route, frequency, duration, prescribed_date, cost

Captures medication orders/administrations tied to an encounter (maps to FHIR medication resources conceptually).

8) claims_and_billings.csv — Charges, claims, payments - MAIN

PK: billing_id

FK:

- patient_id → **patients.patient_id**
- encounter_id → **encounters.encounter_id**
- insurance_provider → **insurers.code**

Non-key: insurance_provider, payment_method, claim_id, claim_billing_date, billed_amount, paid_amount, claim_status, denial_reason

Represents charge/claim lines and adjudication status at a summary level; claim_id links to denial details. The FHIR Claim resource is the analog for reimbursement requests.

9) denials.csv — Denials & appeals lifecycle - MAIN

Columns: claim_id, denial_id, denial_reason_code, denial_reason_description, denied_amount, denial_date, appeal_filed, appeal_status, appeal_resolution_date, final_outcome.

Tracks denial reasons and appeal outcomes per claim. Useful for payer performance and revenue cycle analysis. (Claim/denial processes align with FHIR financial resources.)

10) insurers.csv — Payer dictionary — EXTRA (lookup)

PK: insurer_id

Unique natural key: code (*e.g., UHC, Aetna, BCBS, Humana, Cigna, Medicare, Medicaid*)

Non-key: name, payer_type (*commercial/public*)

Referenced by: patients.insurance_type → insurers.code

Notes: Centralized codebook to standardize payer names/codes.

11) specialty_heads.csv — One head doctor per specialty — EXTRA (organizational/lookup)

PK: head_id

FK (recommended & implemented): head_provider_id → providers.provider_id (*validates that the head is a real provider*)

Non-key: specialty, head_name, head_email

Referenced by: providers.supervisor_provider_id → specialty_heads.head_id

Notes: Ensures a single designated head per specialty for governance and reporting.

Member 1
Signature

Full name

Abdulsamet Ekinci

Student no

150220723

Responsibility:

Managing team – DB Analysis

Tables: patients (**FK:insurance_type**),
encounters (FK: patient_id, provider_id),
insurers(PK)

Member 2
Signature

Full name

Uğur Erkan Bol

Student no

150210074

Responsibility

UI/UX design – Tester

Tables : providers (FK: head_id), diagnoses (FK:
encounter_id), specialty_heads(FK:
head_provider_id)

Member 3
Signature

Full name

Furkan İslamoğlu

Student no

150220065

Responsibility:

Frontend Lead

Tables: **procedures (FK: encounter_id,**
provider_id), medications (FK: encounter_id,
prescriber_id)

Member 3
Signature

Full name

Mustafa Talha Hamsoğlu

Student no

150220066

Responsibility:

Backend Lead

Tables: **calims_and_billing (FK: patient_id,**
encounter_id), denials (FK: claim_id →
billing.claim_id)

Member 4

Signature

Full name

Mustafa Yunus Diler

Student no

150210003

Responsibility:

Backend Optimization

Tables: lab_results (FK: encounter_id)

